TCNOpen	Open Source Developement



TRDP

Train Real Time Data Protocol

TRDP Conformance Test Specification

Document reference no: TCN-TRDP1-A-BOM-031-04

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Document date: 22 December 2023
Revision: 4
Status: issued

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DOCUMENT SUMMARY SHEET

This document contains the TRDP test specification

Participants		
Name and Surname	Organisation	Role
Peter Liesner	NewTec	Participant

History			
V1	23 May 13	Armin-H. Weiss	Initial version
V2	24 May 13	Armin-H. Weiss	Reviewed by UniControls and issued
V3	17 Sept 13	Armin-H. Weiss	License conditions added and issued
V4	22 Dec 23	Peter Liesner	Tests for High Performance and VLAN added and issued

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1. Introduction

1.1. Purpose

This document describes tests verifying the conformance of a TRDP implementation.

1.2. Intended Audience

This document is intended to be used for verification of the TCNOpen TRDP implementation.

1.3. References/Related Documents

Reference	Number	Title
[Wire]	IEC61375-2-3	TRDP Protocol (Annex A)
[Req]	TCN-TRDP1-D-BOM-003	TRDP System Requirement Specification
[TestRep]	TCN-TRDP1-D-BOM-032	TRDP Conformance Test Report

Table 1: References

1.4. Abbreviations and Definitions

Abbreviation	Definition

Table 2: Abbreviations and Definitions

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2. Conformance Tests

The following tests verify the conformance of the TCNOpen TRDP implementation for a specified TRDP Version x.x.x.x (SVN yyyy).

For testing the programs “test/pdpatterns/trdp-pd-test-fast.c” and “test/mdpatterns/trdp-md-test-fast.c” delivered with this TRDP version shall be used as well as trdp-pd-test.c and trdp-md-test.c. The “fast” tests are applied in two queue indexing schemes. One with decimal based message cycle periods and the other with binary based message cycle periods.

For testing with vlan interfaces the program “test/pdpatterns/trdp-pd-test-vlan.c” shall be used.

For documentation of the test results [TestRep] shall be used as template.

2.1. Process Data

PD tests verify the exchange of process data between two devices A and B. All in [wire] defined PD patterns are tested.

All the test cases run continuously in parallel during the test session.

Following table defines the test cases performed on the two platforms and implementations for all tests except pdtest-fast-base-2:

Pattern	Destination	Direction	Data Size in Bytes	Period in ms	Result
PUSH	unicast	A->B, B->A	256	100	
				250	
			1432	100	
				250	
	multicast	A->B, B->A	256	100	
				250	
			1432	100	
				250	
PULL	unicast / unicast	A->B->A, B->A->B	256	500	
			1432	500	
	multicast / multicast	A->B->A, B->A->B	256	500	
			1432	500	

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Table 3: PD Test Patterns

Following table defines the test cases performed on the two platforms and implementations for pdtest-fast-base-2:

Pattern	Destination	Direction	Data Size in Bytes	Period in ms	Result
PUSH	unicast	A->B, B->A	256	128	
				256	
			1432	128	
				256	
	multicast	A->B, B->A	256	128	
				256	
			1432	128	
				256	
PULL	unicast / unicast	A->B->A, B->A->B	256	512	
			1432	512	
	multicast / multicast	A->B->A, B->A->B	256	512	
			1432	512	

All the test cases run on two target platforms - Linux and Windows.

Following table summarizes the test targets used for the specific tests:

	Device A	Device B
PD1	Windows/TCNOpen	Linux/TCNOpen
PD2	Linux/TCNOpen	Windows/TCNOpen

Table 4: PD Test Targets

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2.2. Message Data

MD tests verify the exchange of message data between two devices A and B.

All defined MD patterns are tested on both supported transmission protocols TCP and UDP.

Following table defines the test cases performed on the two platforms and implementations:

Pro-to-col	Pattern	Destina-tion	Re-plies	Direction	Data Size in Bytes	Result			
							Fast-base10	Fast-base2	VLAN
UDP	notify	unicast	0	A->B	64				
					32768				
	request/ reply	unicast/ unicast	1	A->B->A	64				
					32768				
	request/ reply/ confirm	unicast/ unicast	1	A->B->A->B	64				
					32768				
	notify	multicast	0	A->B	64				
					32768				
	request/ reply	multicast/ unicast	1	A->B->A	64				
					32768				
	request/ reply/ confirm	multicast/ unicast	1	A->B->A->B	64				
					32768				
	request/ reply	multicast/ unicast	?	A->B->A	64				
					32768				
	request/ reply/ confirm	multicast/ unicast	?	A->B->A->B	64				
					32768				
TCP	Notify	unicast	0	A->B	64				
					32768				
	request/ reply	unicast	1	A->B->A	64				
					32768				
	request/ reply/ confirm	unicast	1	A->B->A->B	64				
					32768				

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Table 5: MD Test Patterns

All the test cases run on two target platforms - Linux and Windows.

Following table summarizes the tests performed on the two platforms and implementations:

	Device A	**Device B**
MD1	Windows/TCNOpen	Linux/TCNOpen
MD2	Linux/TCNOpen	Windows/TCNOpen

Table 6: MD Test Targets

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